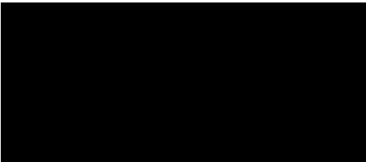




Chemistry Section 1 — Questions



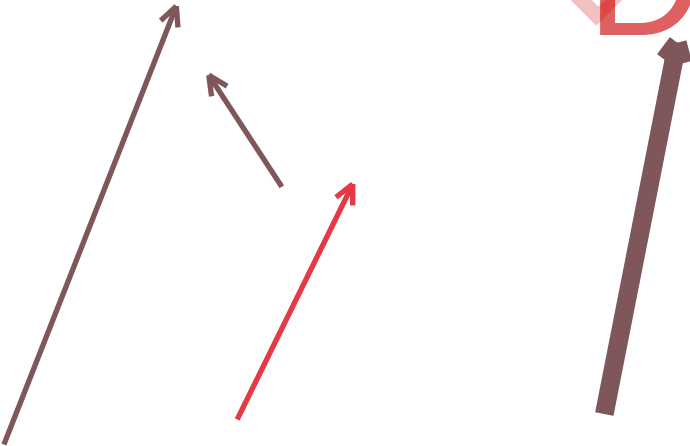
on of Section 1 are given on *page 02* of your question and answer



answer grid on *page 03* of your question and answer booklet.

You may refer to the Chemistry Data Booklet for National 5.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



tihshisdf

SECTION 1 — 25 marks

Attempt ALL questions

1. Oganesson is a newly discovered element with a predicted electron arrangement of 2,8,18,32,32,18,8

In which group of the periodic table should oganesson be placed?

- A 0
B 1
C 2
D 7

 wrong

2. Which line in the table shows the correct number of protons and electrons for the atom, $^{56}_{26}\text{Fe}$?

	Protons	Electrons
A	56	56
B	56	26
C	26	26
D	26	56

3. Which of the following molecules contains three atoms?

- A Phosphorus trichloride
B Magnesium chloride
C Sodium fluoride
D Carbon dioxide

4. Which bonding and structure is never found in elements?

- A Covalent molecular
B Ionic lattice
C Metallic lattice
D Covalent network



5. Which of the following compounds would conduct electricity at 600 °C?

You may wish to use the data booklet to help you.

- A Silicon dioxide
- B Lithium bromide
- C Ammonia
- D Barium chloride

6. Which of the following oxides, when shaken with water, would give an alkaline solution?

You may wish to use the data booklet to help you.

- A Calcium oxide
- B Nitrogen dioxide
- C Sulfur dioxide
- D Nickel oxide

7. A student made some statements about the effect of adding water to a solution of hydrochloric acid.

Identify the correct statement.

- A The rate of reaction with magnesium will increase.
- B The concentration of H^+ ions will increase.
- C The acid will be neutralised.
- D The pH will increase.

8. Ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$, has a gram formula mass of 149.

The percentage by mass of nitrogen in ammonium phosphate is equal to

- A $\frac{14}{149} \times 100$
- B $\frac{42}{149} \times 100$
- C $\frac{149}{14} \times 100$
- D $\frac{149}{42} \times 100$

[Turn over